Rubash Mali
Saxton Calvert
Project 6

Introduction

The Republican River Basin spans Colorado, Kansas, and Nebraska, where increasing irrigation demand and groundwater pumping for agriculture and development have contributed to declining streamflow and interstate water disputes. Understanding how precipitation, evapotranspiration (ET), groundwater pumping, and seasonal patterns influence baseflow is important for water managers responsible for allocating water resources and evaluating the long-term sustainability of river systems, as well as for people living in the region who depend on this vital resource. This project uses multiple linear regression to predict estimated monthly baseflow at streamflow monitoring stations throughout the basin. Our analysis gives water managers a simple, transparent model of how pumping, ET, and precipitation relate to baseflow at each station, helping them see which factors matter most when planning water use. The goal is to determine which factors best explain baseflow variation and whether a relatively simple statistical model can provide useful predictions for water-resource planning.

We compared ten progressively complex regression models, including an average-only baseline, using observations from 1939–1989 for training and 1990–2000 for testing. Our best model predicts log-transformed baseflow using station identity, seasonal effects, environmental inputs, and a year trend. It achieved a Nash–Sutcliffe efficiency (NSE) of 0.798 and a percent bias (PBIAS) of +3.36%. The results indicate that differences between monitoring stations and the long-term decline in baseflow are particularly important for prediction. While these findings are not definitive enough to guide water allocation on their own, they provide a useful starting point for understanding baseflow patterns.

Repository: https://github.com/loudpugpug42/project6
Slides: Presentation

Dataset

The dataset was provided for the course and is based on the Republican River data analyzed by Xu and Valocchi (2015). It contains 15,591 monthly observations from 46 station locations (42 segment IDs) between 1939 and 2000. Variables include estimated baseflow, precipitation, ET, groundwater pumping (stored as negative values since water is being removed), station location, and time. Baseflow represents the portion of streamflow attributed to groundwater discharge. It is estimated by separating groundwater contributions from total streamflow, so it is not a direct measurement, and the dataset does not state its units. The dataset captures seasonal variation, differences among rivers and creeks, and long-term changes associated with water use and environmental conditions.

Exploratory data analysis revealed important issues in the dataset. Nine stations had artificial zero values across their input variables, and some segment identifiers represented multiple

stations. Precipitation was recorded annually rather than monthly, with the same yearly total repeated in every month. After removing the nine stations, 11,254 rows from 37 stations remained. These still contained 232 negative baseflow estimates, almost all at one station (segment 178), which required special treatment during log transformation. The cleaning steps are described in the Technical section.

Analysis Technique

We used multiple linear regression because it estimates how several environmental and temporal variables relate to baseflow simultaneously, and its coefficients are easy to interpret. We built a sequence of models, or "model ladder," where each model builds off the previous ones. This allowed us to evaluate the contribution of individual features before comparing them with a model containing a larger set of predictors. Because rivers differ substantially in size and their seasonal fluctuations are more proportional than absolute, we also tested regression on log-transformed baseflow. This transformation improved the model's ability to represent relative changes across stations.

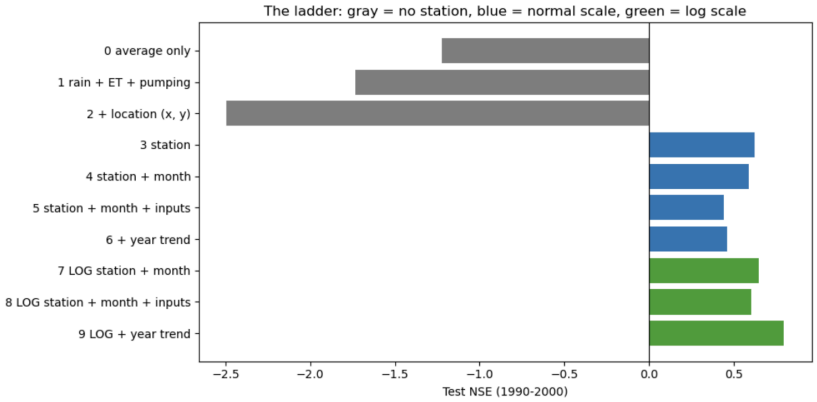
We split the data by year instead of randomly. Neighboring months are very similar, so a random split would let the model train on almost the same data it is tested on. A time-based split better matches the real task of predicting future conditions from past observations. We scored models using NSE, MAE, and PBIAS, the same metrics used by Xu and Valocchi (2015). An NSE of 1 is a perfect fit, 0 means the model is no better than predicting the average, and below 0 means it is worse.

Results

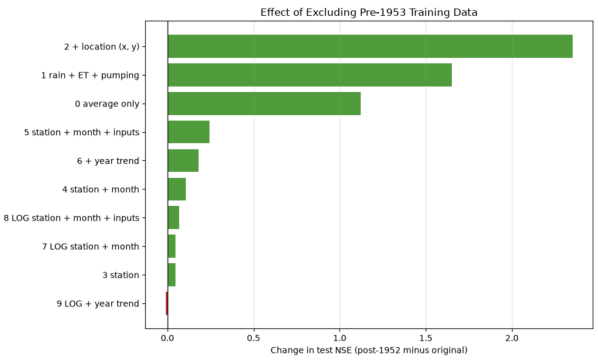
Station identity was the most important starting point for prediction. Models using precipitation, ET, pumping, or geographic coordinates without station identity performed worse than a simple average baseline, demonstrating that environmental inputs alone could not explain the large differences between rivers and creeks. Adding station identity raised test NSE to 0.623, while the log-transformed model with station and month effects reached 0.650. This would suggest to water managers that the locations for best baseflow would be those stations with higher estimates, and the places or people requiring more such as farmers of water-intensive crops would be best suited to go there for more water. Our best model added environmental inputs and a year trend to the log-scale regression. It achieved an NSE of 0.798, a mean absolute error (MAE) of 4.873 in the dataset's unspecified baseflow units, and a PBIAS of +3.36%, indicating slight overall underprediction. Its estimated coefficients were

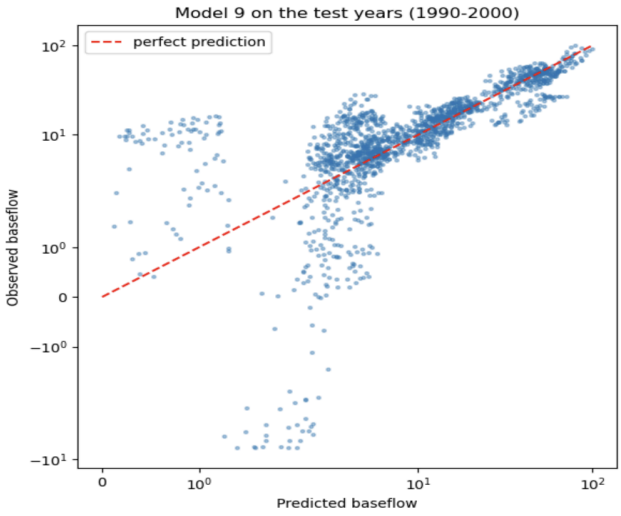
model	parameters	train_NSE	test_NSE	test_MAE	test_PBIAS	DW
0 average only	0	0.000	-1.224	23.145	-110.961	0.047
1 rain + ET + pumping	3	0.035	-1.734	24.317	-107.033	0.054
2 + location (x, y)	5	0.053	-2.493	27.201	-118.295	0.053
3 station	35	0.800	0.623	6.475	-19.921	0.254
4 station + month	46	0.809	0.590	7.797	-20.630	0.251
5 station + month + inputs	49	0.813	0.441	9.046	-21.270	0.261
6 + year trend	50	0.814	0.461	8.855	13.162	0.263
7 LOG station + month	46	0.828	0.650	6.095	-16.347	0.249
8 LOG station + month + inputs	49	0.856	0.602	6.380	-18.634	0.280
9 LOG + year trend	50	0.856	0.798	4.873	3.360	0.288

consistent with hydrologic expectations: greater ET and increased pumping were associated with lower baseflow while greater precipitation with higher baseflow. The year trend corresponded to an estimated baseflow decline of approximately 0.8% annually. Adding many more predictors produced little improvement, suggesting that our model captured most of the predictive information available in the supplied features. This would help water managers within the basin to know what actions to take that would help increase baseflow within certain parts of the region and how to reduce it should flooding or similar events occur.



The model's performance was comparable, though not directly equivalent, to the published study of the Republican River Basin. That study reported an NSE of 0.83 for the RRCA groundwater model alone and 0.93 after applying random forest or support vector regression corrections. However, residuals remained strongly correlated across consecutive months, and several stations were substantially over- or underpredicted. In a historical sensitivity analysis, restricting training to 1953–1989 due to the completion of the Harlan County Dam changed the model's test NSE from 0.798 to 0.787 and MAE from 4.873 to 4.963, while improving PBIAS from +3.36% to +0.48%. Excluding pre-dam observations slightly reduced predictive accuracy but reduced overall bias. These results suggest that earlier observations contain useful predictive information; they do not establish that the dam caused the differences. Overall our best performing model, while requiring consideration of its errors and assumptions, provides a useful baseline for exploring baseflow patterns and related variables for water-management decisions made by officials with regards to the Republican River Basin.





Technical

Initial analysis found nine stations with artificial zeros for all input variables, about 28% of rows, leading to their exclusion. We used a combined station identifier (segment ID plus x-coordinate) instead of segment ID because some segments represented multiple stations. The date formula provided required correcting month labels by shifting dates forward 16 days. We did our train/test split by year, with training data from 1939–1989 and testing data from 1990–2000. For log-scale models, negative baseflow estimates were set to zero and we modeled log(baseflow + 1), which keeps zero values defined. Predictions were converted back to the original scale (exp(prediction) − 1) before calculating evaluation metrics. Because this treatment loses information about negative values, results for affected stations (mainly segment 178) require caution. We also repeated the model comparison using only 1953–1989 training observations to investigate sensitivity to the period before the Harlan County Dam was filled.

We evaluated a ladder of ten models (Models 0–9) of increasing complexity. We began with an average-only baseline and added environmental inputs, location, station identity, seasonal effects, log transformation, and a year trend. Test performance was assessed using NSE, MAE, and PBIAS, with the Durbin–Watson statistic used to examine correlation in residuals. The first models showed that environmental variables and coordinates alone were insufficient. It became clear that station identity produced the largest improvement. Adding month and environmental predictors on the original scale reduced performance, whereas log transformation and the year trend improved predictions.

Model 9 was selected because it offered the best overall test performance. Adding a substantially larger feature set (cumulative pumping, previous-year precipitation and pumping, and station-specific pumping effects) yielded little additional benefit. Diagnostic plots showed uneven residual spread, with larger errors for small creeks, and heavy-tailed errors. A Durbin–Watson statistic of approximately 0.29 indicated strong temporal dependence, so our regression p-values may be overly optimistic. Alternative approaches considered include station-specific or regularized regression, explicit time-series error modeling, and incorporating previous-month baseflow when available at prediction time. The post-1952 comparison also showed that changing the training period alone did not improve the best model, so the dam's influence would require a more targeted analysis if it were to be pursued.